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1677	

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Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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Office Action Summary	Application No. 18/283,274	Applicant(s) Kobayashi et al.	
	Examiner ELIZABETH OGUNTADE	Art Unit 1677	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 19 March 2026.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-14 is/are pending in the application.

5a) Of the above claim(s) 9-14 is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-8 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☐ The drawing(s) filed on _____ is/are: a)☐ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date 12/21/2023, 01/19/2024, 04/24/2024, 10/04/2024

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

Notice of Pre-AIA or AIA Status

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Election/Restrictions

Applicant's election without traverse of **Group I (claims 1-8)**, drawn to a method for assisting in the determination of malignant pancreatic cystic tumor by measuring APOA2-AT and/or APOA2-ATQ proteins and applying malignancy discrimination criteria, in the reply filed on **03/19/2026** is acknowledged. Hence, claims **9-14** are withdrawn from further consideration pursuant to 37 CFR 1.142(b) as being drawn to a nonelected **invention**, there being no allowable generic or linking claim.

Status of the Claims

Claims 1-14 are pending. Claims 9-14 are withdrawn. Claims 1-8 are examined herein in view of the restriction.

Priority

The present application, filed 09/21/2023, is a 371 of PCT/JP2022/013234, filed 03/22/2022, which claims foreign priority of JP2021-049931, filed 03/24/2021. Should applicant desire to obtain the benefit of foreign priority under 35 U.S.C. 119(a)-(d) prior to declaration of an interference, a certified English translation of the foreign application must be submitted in reply to this action. 37 CFR 41.154(b) and 41.202(e).

Failure to provide a certified translation may result in no benefit being accorded for the non-English application.

Information Disclosure Statement

The Information Disclosure Statement(s) filed 09/21/2023, 12/21/2023, 01/19/2024, 04/24/2024, and 10/04/2024 are acknowledged and have been considered.

Claim Rejections - 35 USC § 112

The following is a quotation of the first paragraph of 35 U.S.C. 112(a):

> (a) IN GENERAL.—The specification shall contain a written description of the invention, and of the manner and process of making and using it, in such full, clear, concise, and exact terms as to enable any person skilled in the art to which it pertains, or with which it is most nearly connected, to make and use the same, and shall set forth the best mode contemplated by the inventor or joint inventor of carrying out the invention.

The following is a quotation of the first paragraph of pre-AIA 35 U.S.C. 112:

> The specification shall contain a written description of the invention, and of the manner and process of making and using it, in such full, clear, concise, and exact terms as to enable any person skilled in the art to which it pertains, or with which it is most nearly connected, to make and use the same, and shall set forth the best mode contemplated by the inventor of carrying out his invention.

Claims **1-8** are rejected under 35 U.S.C. 112(a) or 35 U.S.C. 112 (pre-AIA), first paragraph, **because the specification, while being enabling for** measuring APOA2-AT and APOA2-ATQ proteins in a body fluid sample and demonstrating certain embodiments of distinguishing benign and malignant intraductal papillary mucinous neoplasm (IPMN) based on measured amounts as shown in the specification, **does not reasonably provide enablement for** assisting in the determination of malignant pancreatic cystic tumor broadly, including across the full scope of pancreatic cystic tumors and all implementations of determining benign versus malignant "on the basis of the amount" of APOA2-AT and/or APOA2-ATQ as recited in claims 1–8. The specification does not enable any person skilled in the art to which it pertains, or with which it is most nearly connected, to **use** the invention commensurate in scope with these claims.

The enablement requirement is such that the specification of the described invention in such terms that one skilled in the art can make and use the invention to ensure that the invention is communicated to the interested public in a meaningful way. (MPEP 2164)

The standard for determining whether the specification meets the enablement requirement is determined in view of the Wands factors (MPEP 2164.01(a)), to assess whether any necessary experimentation required by the specification is "reasonable" or is "undue." The factors to be considered in determining whether undue experimentation is required include: (1) the quantity of experimentation needed to make or use the invention based on the content of the disclosure, (2) the amount of direction provided by the inventor, (3) the existence of working examples, (4) the nature of the invention, (5) the state of the prior art, (6) the level of one of ordinary skill, (7) the level of predictability in the art, and (8) the breadth of the claims. While all of these factors are considered a sufficient amount for a prima facie case are discussed below.

The bread of the claims

Claim 1 broadly recites a method for assisting in determining whether a pancreatic cystic tumor is benign or malignant by measuring APOA2-AT and/or APOA2-ATQ in a body fluid sample and using the amount for that determination. This is not limited to any specific tumor subtype and therefore covers all pancreatic cystic tumors. The “and/or” language further expands the scope to include use of either marker alone or both together. The claim also does not require any specific cutoff, threshold, or decision rule, and thus encompasses any manner of interpreting biomarker levels. Dependent claims 2–8 do not meaningfully narrow this breadth. Claim 2 limits the tumor to IPMN or MCN but still covers all implementations, while claims 3–6 add general directional or cutoff concepts without defining specific, universally applicable criteria. Claim 7 only limits the sample type, and claim 8 adds assay steps and a discriminant but still broadly applies to pancreatic cystic tumors without limiting how the determination

is made. Accordingly, the claims cover all methods of using APOA2-AT and/or APOA2-ATQ levels across all pancreatic cystic tumors and analytical approaches, making them significantly broader than the embodiments disclosed in the specification.

The nature of the invention

The nature of the invention is a biomarker-based diagnostic method that uses measured levels of specific protein isoforms, namely APOA2-AT and APOA2-ATQ, in a body fluid sample to assist in determining whether a pancreatic cystic tumor is benign or malignant. The invention is not merely directed to detecting the presence of these proteins, but to interpreting their measured amounts to make a clinical determination about tumor status.

The amount of direction provided by the inventor/ The existence of working examples

The specification provides working examples and some direction, but the disclosure is limited and does not reasonably enable all implementations of determining benign versus malignant pancreatic cystic tumor “on the basis of the amount” across the full scope of the claims.

The specification includes experimental work in Example 3, where plasma samples from patients with IPMN are analyzed (paragraph [0130], page 58). The example further describes measuring APOA2-AT protein in plasma using a sandwich ELISA with a specific anti-APOA2-AT terminus polyclonal antibody and an anti-APOA2 non-terminus monoclonal antibody (paragraph [0131], page 58), and the example describes measuring APOA2-ATQ protein using a corresponding anti-APOA2-ATQ terminus monoclonal antibody and an anti-APOA2 non-terminus polyclonal antibody (paragraph [0132], page 59). The results of these measurements are presented in Figure 4 (APOA2-AT concentration in blood for benign IPMN and malignant IPMN) and Figure 5 (APOA2-ATQ concentration in blood for benign IPMN and malignant IPMN). Additionally, the example explains that a box-and-whisker plot is prepared using the measured values - a

mean of the APOA2-AT protein concentrations was found lower in the malignant group compared with the benign group, and a mean of the APOA2-ATQ protein concentrations was found higher in the malignant group compared with the benign group (paragraph [0133], page 60). Then, in Example 4, the specification further describes ROC analysis, and Figure 7 depicts AUC values of 0.762 for APOA2-AT and 0.648 for APOA2-ATQ (paragraph [0142], page 63). These disclosures demonstrate that the specification provides working examples showing some degree of discrimination between benign and malignant IPMN based on biomarker amounts.

The specification also provides direction for performing the claimed method. Specifically, it describes a sequence of steps including measuring the amount of APOA2-ATQ protein using a terminus antibody and a non-terminus antibody, measuring the amount of APOA2-AT protein using corresponding antibodies, and inputting the measured values into a preset discriminant to compare with a discriminant value (paragraph [0012], pages 7-8). Moreover, the specification further describes that malignancy can be determined when the measurement value of the APOA2-AT protein in the test subject is equal to or lower than the cutoff value set on the basis of the amount of the APOA2-AT protein in a known benign pancreatic cystic tumor patient group (paragraph [0103], page 48), or when the measurement value of the APOA2-ATQ protein in the test subject is equal to or higher than the cutoff value set on the basis of the amount of the APOA2-ATQ protein in a known benign pancreatic cystic tumor patient group (paragraph [0104], page 49).

However, the specification does not reasonably provide enablement for all implementations of determining benign versus malignant “on the basis of the amount” as broadly recited in the claims. First, the working examples are limited to IPMN, as explicitly described in paragraph [0130], and all corresponding figures (Fig. 4, Fig. 5, Fig. 7) relate only to benign and malignant IPMN. The specification does not provide working examples for other pancreatic cystic tumor types encompassed by the claims, such as MCN or other cystic lesions. Second, although the specification states in paragraphs [0103] –

[0104] that malignancy may be determined based on lower or higher amounts or cutoff values, it does not provide a generalized rule, standardized threshold, or validated discriminant applicable across different tumor types or patient populations, but instead relies on comparisons to "known benign pancreatic cystic tumor", "known malignant pancreatic cystic tumor," or cutoff values derived from limited data. Third, the experimental results themselves, as shown in Figures 4 and 5, present overlapping distributions between benign and malignant groups, and the ROC values reported in paragraph [0142] (AUC of 0.762 and 0.648) indicate only moderate discriminatory ability, which does not establish that biomarker amount alone can be reliably used across all embodiments to determine benign versus malignant status.

Accordingly, while the specification provides specific working examples and some procedural direction, those examples are confined to a narrow experimental context and do not teach how to apply the claimed method across the full breadth of pancreatic cystic tumors or across all claimed implementations of determining benign versus malignant "on the basis of the amount." A person of ordinary skill in the art would therefore not be able to use the invention commensurate in scope with the claims without undue experimentation.

The state of the prior art/the level of predictability in the art

The state of the prior art and the level of predictability in the art weigh against enablement, because the prior art shows that APOA2-based biomarker testing was understood in the art primarily as a tool for screening, risk stratification, and detection of pancreatic cancer or high-risk status, not as a predictable and generally applicable method for determining whether a pancreatic cystic tumor is benign or malignant solely on the basis of the amount of APOA2-AT and/or APOA2-ATQ across the full scope now claimed. The prior art also shows that the biology of APOA2 isoforms was complex, that signal behavior overlapped across multiple pancreatic and non-pancreatic disease states, and that more elaborate models

were often needed to obtain meaningful clinical utility. Taken together, the art did not present a simple, settled, and predictable relationship between APOA2 marker amount and benign-versus-malignant cystic tumor status.

First, the Honda et al. (Plasma Biomarker for Detection of Early Stage Pancreatic Cancer and Risk Factors for Pancreatic Malignancy Using Antibodies for Apolipoprotein-AII Isoforms. Scientific Reports. Vol. 5, No. 1, November 2015 – IDS dated 09/21/2023), shows that the field understood apoAII-ATQ/AT as a marker for early pancreatic cancer detection and identification of patients at high risk for pancreatic malignancy, not as a standalone benign-versus-malignant cystic tumor classifier. In particular, Honda et al. states that apoAII-ATQ/AT might serve as plasma biomarkers for the early detection of pancreatic cancer and that the marker also identified patients at high risk for pancreatic malignancy (Abstract, page 1). Honda et al. further discloses that plasma/serum biomarkers for the early detection of pancreatic cancer would be useful clinical tools for screening patients in order to identify those who should undergo a second screening using stricter diagnostic modalities that can detect pancreatic dysfunction before imaging (paragraph 1, page 2), and further describes development of sandwich ELISAs for apoAII isoforms and evaluation of samples from patients with pancreatic cancer, pancreatic disorders including precancerous lesions, and other malignant diseases (paragraph 4, page 2). That is crucial because it shows that the art positioned the biomarker as part of a screening workflow, not as a definitive classifier for benign versus malignant pancreatic cystic tumor.

Moreover, Honda et al. also teaches that apoAII isoform behavior was biologically heterogeneous and mechanistically uncertain, which cuts against predictability. Specifically, Honda et al. discloses that the mass spectrometry (MS) data suggested two different types of IDACP patients, one classified by hyper-processing of the apoAII homodimer and the other by hypo-processing, and then states that a formula had to be developed to calculate a surrogate biomarker level of apoAII-ATQ/AT from ELISA results for apoAII-ATQ and apoAII-AT, with the solution of that equation being defined as the surrogate biomarker

(paragraph 1, page 4). Honda et al. further states that the mechanism by which the C-terminus of the circulating apoAII homodimer is modified was unclear, that hypo- and hyper-processing patterns were characteristic of IDACP or other pancreatic disorders, and that their results suggested reduced apoAII-ATQ/AT in patients with IDACP and other pancreatic disorders under conditions of hyper- or hypo-processing (paragraphs 3-4, page 11). Additionally, Honda et al. further discloses that direct absolute quantitation of apoAII-ATQ/AT was not achieved, that equation-1 could not predict an absolute quantity of apoAII-ATQ/AT, and that the ELISA outputs for apoAII-ATQ and apoAII-AT were instead used as surrogate biomarkers (paragraph 1, page 12). This prior art therefore demonstrates that even for pancreatic cancer detection, the marker system was not a simple direct measurement with straightforward interpretation.

Lastly, Honda et al. further shows that the marker was not specific to malignant cystic tumors and instead changed across many pancreatic and non-pancreatic conditions. Specifically, Honda et al. discloses that apoAII-isoforms cannot be used to distinguish IDACP from other pancreatic diseases, and that alterations in the distribution of apoAII-isoforms are associated with not only IDACP but also other pancreatic disorders, such as endocrine tumors of the pancreas, IPMN, MCN, SCN, and chronic pancreatitis (paragraph 5, page 12). This disclosure is directly relevant here: if the prior art itself says the biomarker changes across many pancreatic disorders and cannot distinguish pancreatic cancer from other pancreatic diseases, then the art was not predictable with respect to using biomarker amount alone to sort benign versus malignant pancreatic cystic tumors across a broad genus.

Second, Sanda et al. (EP3054298B1) is likewise highly relevant because it demonstrates that APOA2-variant testing was developed as a method for detecting pancreatic tumor broadly, including pancreatic cancer and benign pancreatic tumor, but that doing so required more structured assay formats, discriminants, and even follow-on cancer-marker testing. Specifically, Sanada et al. discloses that conventional tumor markers used in clinical diagnosis had major problems because the great majority

offered only a positive rate on the order of 50 to 70% and most exhibited false-negative results, particularly for early cancers (paragraph [0002], page 3). Here the prior art recognized that the field was difficult and underdeveloped. Furthermore, Sanada et al. demonstrates that that the marker system required technical restructuring to obtain useful information. Particularly, Sanada et al. teaches that APOA2-ATQ/AT protein dimer was significantly decreased in pancreatic cancer patients compared with normal persons and that pancreatic cancer could be detected with accuracy as high as an AUC value of 0.85 or higher using that protein dimer as a pancreatic cancer marker (paragraph [0008], page 4). However, Sanada et al. explains that this required complicated mass spectrometry workflows and that the approach faced problems including susceptibility of reagents, sensitivity of capture efficiency to washing and reagent preparation, low throughput, signal interference among proteins, difficulty of signal attribution, and challenges in quantitative performance, making it unsuitable for diagnostic purposes requiring highly accurate measurement (paragraph [0009], page 4).

Moreover, Sanada et al. reveals that that the inventors first attempted to detect the APOA2-ATQ/AT dimer directly using antibodies against the respective C-terminal regions of APOA2-ATQ and APOA2-AT, but that the dimer could not be detected with high accuracy, probably due to interfering substances in blood or steric hindrance caused by the two antibodies binding to the same antigen (paragraph [0015], page 5). Sanada et al. further discloses that they instead separately measured total amount of APOA2-ATQ protein and total amount of APOA2-AT by sandwich ELISA using terminus and non-terminus antibodies in combination, obtained the two measurement values, and combined the results; then, further states that pancreatic cancer patients could be discriminated from normal persons with high accuracy and that benign pancreatic tumor could also be detected using this technique (paragraph [0016], page 5). This prior art therefore confirms that the art did not regard the problem as simple or predictable; rather, it required reformulation of the assay concept and combination analysis.

Lastly, for predictability, Sanada et al. shows that the art used preset discriminants and additional markers, not simple universal amount rules. In paragraph [0017], especially aspects (1), (3), (4), (5), and (7), defines a detection method in which APOA2-ATQ and APOA2-AT are measured separately, the resulting values are input into a preset discriminant, the discriminant may be a logistic regression expression, support vector machine expression, neural network expression, or discriminant analysis expression, the logistic regression may use APOA2-ATQ, APOA2-AT, and/or their product as variables, and a fourth step may measure CA19-9 or DU-PAN-2 in a test subject already determined to have pancreatic tumor, in order to determine the test subject to have pancreatic cancer when the measurement value exceeds a predetermined reference value and determining the test subject to have benign pancreatic tumor when the measurement value is equal to or lower than the reference value (page 5). Paragraph [0018] then states that particular APOA2 protein variants in blood can be merely measured to determine whether a patient has a pancreatic tumor or to evaluate the possibility of having a pancreatic tumor (page 6). Paragraphs [0107]–[0112] similarly explain that the determining step involves measuring APOA2-ATQ and APOA2-AT, inputting both values into a preset discriminant, and optionally using a known pancreatic cancer marker in a further step to discriminately determine pancreatic cancer versus benign pancreatic tumor (page 19). This is crucial evidence that the prior art did not treat the field as predictable enough that one could generally decide benign versus malignant status from “amount” alone across a broad cystic tumor genus. Instead, the art taught model-based analysis and confirmatory marker stratification.

Accordingly, when Honda et al. and Sanada et al. are considered together, they show that the state of the prior art was one of screening-oriented biomarker development with substantial biological heterogeneity, incomplete mechanistic understanding, overlapping abnormal results across multiple pancreatic and non-pancreatic conditions, and dependence on surrogate calculations or preset discriminants. That evidentiary record supports a finding that the level of predictability in the art was low, especially for the presently claimed subject matter of assisting in the determination of whether a

pancreatic cystic tumor is benign or malignant broadly on the basis of the amount of APOA2-AT and/or APOA2-ATQ. This Wands factor therefore weighs in favor of non-enablement.

The level of ordinary skill in the art

The level of ordinary skill in the art at the time of the effective filing date would have been relatively high, encompassing individuals with advanced training and multidisciplinary expertise in clinical diagnostics, proteomics, and pancreatic disease biology. However, even at this high level of skill, the prior art demonstrates that a person of ordinary skill would not have had a reasonable expectation of being able to practice the full scope of the claimed invention without undue experimentation, particularly for distinguishing benign versus malignant pancreatic cystic tumors based solely on APOA2 biomarker amounts. A person of ordinary skill in this field would likely have had at least a Ph.D., M.D., or equivalent advanced degree in a relevant discipline such as biochemistry, molecular biology, clinical pathology, or oncology, along with several years of experience in biomarker discovery, immunoassay development (e.g., ELISA), and/or pancreatic disease diagnostics. This is supported by the sophistication of the techniques disclosed in the prior art. Therefore, while the level of ordinary skill in the art was high, this factor does not weigh in favor of enablement. Instead, it underscores that even highly skilled practitioners would have needed substantial experimentation, model development, and clinical validation to practice the full scope of the claimed invention, particularly with respect to reliably distinguishing benign from malignant pancreatic cystic tumors using APOA2 biomarker levels alone.

The quantity of experimentation needed to make or use the invention based on the content of the disclosure

The quantity of experimentation required to make and use the full scope of the claimed invention, based on the content of the disclosure, would be extensive and undue, weighing against enablement. Practicing the full scope of the claims would require extensive experimentation. The specification provides data only for IPMN, requiring additional studies for other tumor types. The specification does not provide universal cutoff values or decision rules, requiring further experimentation to establish thresholds and validate diagnostic performance. The overlapping distributions shown in Figures 4 and 5 and the moderate AUC values in Figure 7 indicate that additional optimization would be required. Furthermore, implementing the preset discriminant recited in claim 8 would require developing and validating statistical models across diverse datasets. The prior art confirms that such models require substantial data collection and validation. These efforts would involve extensive clinical studies and iterative refinement, constituting undue experimentation.

Therefore, considering the Wands factors as a whole, the specification enables only a limited subset of embodiments, primarily involving APOA2 biomarker measurement and discrimination within a specific IPMN dataset. The claims, however, encompass a much broader scope, including all pancreatic cystic tumors and all implementations of determining benign versus malignant status based on biomarker amount. Given the breadth of the claims, the limited working examples, the lack of generalizable guidance, the unpredictability of the art, and the substantial experimentation required, claims 1-8 are rejected under 35 U.S.C. 112(a)

Claim Rejections - 35 USC § 101

35 U.S.C. 101 reads as follows:

> Whoever invents or discovers any new and useful process, machine, manufacture, or composition of matter, or any new and useful improvement thereof, may obtain a patent therefor, subject to the conditions and requirements of this title.

Claims **1-8** are rejected under 35 U.S.C. 101 because the claimed invention is directed to a judicial exception, specifically a natural law (i.e., a naturally occurring correlation between APOA2 isoform levels and tumor malignancy) and an abstract idea (i.e., the mental process of evaluating and interpreting the measured data to reach a diagnostic conclusion).

This rejection is made in accordance with Patent Subject Matter Eligibility as set forth in MPEP §2106. Analysis of subject-matter eligibility under 35 U.S.C. §101 requires consideration under these steps as followed:

Step 1 – Statutory Category (Refer to MPEP §2106.03): Claim 1-8 are drawn to a process, which falls within a statutory category under 35 U.S.C. §101.

Step 2A, Prong One – Recitation of a Judicial Exception (Refer to MPEP §2106.04): Regarding claim 1, it recites the following: a method comprising the steps of measuring *in vitro* the amount of APOA2-AT protein and/or APOA2-ATQ protein in a body fluid sample obtained from a test subject, and assisting in determining whether a pancreatic cystic tumor is benign or malignant based on the measured amount. The claim therefore relies on a naturally occurring relationship, namely, the correlation between the levels of specific APOA2 isoform proteins in a biological sample and the presence or absence of malignancy in pancreatic cystic tumors. This relationship is a naturally occurring biological principle that exists in the human body independent of any human action. The measurement steps merely detect or observe these naturally occurring protein levels, and the subsequent "assisting in determining" step applies this natural

relationship to reach a diagnostic conclusion. Accordingly, claim 1 recites a judicial exception in the form of a law of nature/natural phenomenon. Additionally, claim 1 includes steps of evaluating the measured protein amounts and using those values to assist in determining tumor status. Such evaluation and interpretation of data can be performed mentally or with basic tools such as pen and paper once the data is obtained. Therefore, the claim also recites an abstract idea in the form of a mental process (i.e., analyzing information and making a diagnostic determination based on that information). As such, claim 1 recites both a natural law (the biomarker-disease correlation) and an abstract idea (the mental evaluation of the data).

Also, dependent claims 2–8 incorporate all limitations of claim 1. Therefore, they inherit the judicial exceptions recited in claim 1, including the natural law (the correlation between APOA2 isoform levels and malignancy) and the abstract idea (the mental process of evaluating and interpreting the measured data to reach a diagnostic conclusion). The additional limitations in these dependent claims, such as specifying particular tumor types (e.g., IPMN or MCN), defining relative comparisons or cutoff values, or identifying the type of body fluid sample, do not remove or alter the underlying reliance on the natural correlation or the mental evaluation of data.

Regarding claims 3-6, they further recite limitations directed to comparing measured biomarker levels, including determining whether the amount of APOA2-AT protein is lower than a reference value (claim 3) and/or whether the amount of APOA2-ATQ protein is higher than a reference value (claim 5), and in some embodiments comparing such values to predetermined cutoff thresholds (claims 4 and 6). These limitations constitute an abstract idea in the form of a mental process, namely, evaluating, comparing, and interpreting data to reach a diagnostic conclusion. Such comparison and evaluation steps can be performed in the human mind or using basic tools once the data is obtained. Accordingly, claims 3–6 recite judicial exceptions in the form of both the natural law identified in claim 1 (the correlation

between APOA2 isoform levels and tumor malignancy) and an abstract idea (mental processes involving comparison and evaluation of data).

Regarding claim 8, although it depends from claim 1, it independently recites additional steps including: (A) measuring the amount of APOA2-ATQ protein using specific antibodies that bind to defined terminal regions; (B) measuring the amount of APOA2-AT protein using specific antibodies that bind to defined terminal regions; and (C) determining the presence or absence of a pancreatic cystic tumor by inputting the measured values into a preset discriminant and comparing the resulting value to a reference discriminant. Despite these additional elements, claim 8 still fundamentally relies on the same natural relationship between APOA2 isoform levels and tumor malignancy. The antibody-based measurements are merely conventional techniques for detecting the naturally occurring proteins, and the discriminant analysis step applies the detected values to the same underlying correlation. Furthermore, the step of "inputting" measured values into a "preset discriminant" and "comparing" the resulting value constitutes a mathematical concept and/or a mental process. Specifically, this step involves processing numerical data using a decision rule or mathematical model (i.e., a discriminant function) and comparing results to determine a classification outcome. Such operations can be performed mentally or using routine computational tools and therefore fall within the category of abstract ideas, including mathematical concepts and mental processes. Accordingly, claim 8 independently recites judicial exceptions in the form of both a natural law (the biomarker-disease correlation) and an abstract idea (mathematical/mental processing of data to reach a diagnostic determination).

Step 2A, Prong Two – Integration into a Practical Application (Refer to MPEP §2106.04 (d)): Regarding claim 1, the additional elements of claim 1 include, for example: (i) obtaining a body fluid sample from a test subject; (ii) measuring the amount of APOA2-AT protein and/or APOA2-ATQ protein present in the sample; and (iii) using the measured amount to assist in determining whether a pancreatic

cystic tumor is benign or malignant. When considered individually and in combination, these elements do not integrate the identified judicial exceptions into a practical application. First, the step of obtaining a body fluid sample and measuring protein levels constitutes mere data acquisition and detection of naturally occurring substances. These steps are routinely performed in the field of clinical diagnostics using well-understood laboratory techniques such as immunoassays, and therefore represent insignificant extra-solution activity that merely gathers data for use in applying the natural law. Such data-gathering steps do not impose any meaningful limit on the judicial exception. Second, the step of "assisting in determining" whether a tumor is benign or malignant merely applies the natural correlation and abstract idea identified in Step 2A, Prong 1. This step does not recite any particular manner of performing the determination beyond generally using the measured values. It does not require a specific algorithm, thresholding technique, or technological implementation that would meaningfully limit the claim. Instead, it broadly encompasses any use of the correlation to inform a diagnostic conclusion, which is equivalent to merely instructing to apply the judicial exception.

Third, claim 1 does not recite any improvement to an existing technology or technical field. There is no indication that the claimed method improves the functioning of a laboratory assay, enhances detection sensitivity, or introduces a new analytical technique. The claim also does not recite a particular machine or apparatus that is integral to performing the claimed method, but instead relies on generic and unspecified laboratory techniques. Fourth, claim 1 does not effect a transformation of a particular article into a different state or thing beyond the routine handling and analysis of a biological sample. The sample is merely analyzed to extract information (protein levels), which is insufficient to constitute a meaningful transformation under 35 U.S.C. 101. Lastly, claim 1, does not apply the judicial exception to effect a treatment or prophylaxis for a disease or medical condition. Rather, it is limited to providing diagnostic information, which, is insufficient to render the claim eligible. Accordingly, when viewed as a whole, claim

1 merely links the use of the judicial exceptions to the field of medical diagnostics and amounts to no more than a drafting effort designed to monopolize the natural correlation and associated abstract idea.

Moreover, the additional limitations recited in claims 2–8 do not integrate the identified judicial exceptions into a practical application. These claims merely add further detail or context to the method of claim 1, such as specifying particular types of pancreatic cystic tumors (e.g., intraductal papillary mucinous neoplasm (IPMN) or mucinous cystic neoplasm (MCN)), defining how measured values are compared (e.g., relative comparisons between subject groups or comparisons to predetermined cutoff values), or identifying the type of biological sample (e.g., blood, plasma, or serum). These additional elements do not impose any meaningful limitation on the judicial exception. Instead, they merely refine the context, data interpretation, or field of use in which the natural law and abstract idea are applied. For example, specifying a particular tumor type or sample type merely narrows the population or environment in which the method is performed, which is insufficient. Similarly, reciting comparisons to cutoff values or relative levels constitutes conventional data analysis techniques, which remain part of the abstract idea and do not integrate the exception into a practical application. None of these additional limitations improve a technological process, require a particular machine integral to the claim, or effect a meaningful transformation. Accordingly, claims 2–8, when considered individually and in combination with claim 1, fail to integrate the judicial exceptions into a practical application.

Regarding claims 3-6, the additional limitations do not integrate the identified judicial exceptions into a practical application. The recited comparison steps merely apply the natural correlation using conventional data evaluation techniques, such as determining whether a value is higher or lower than a reference or falls within a threshold range. These limitations do not improve any technology or technical field, do not require a particular machine integral to the claim, and do not effect a transformation of matter. Rather, they constitute data analysis and interpretation, which amounts to insignificant extra-

solution activity and merely links the judicial exceptions to the field of medical diagnostics. Accordingly, claims 3–6 do not integrate the judicial exceptions into a practical application.

Regarding claim 8, even when considered independently, the additional elements include: (i) measuring APOA2-ATQ and APOA2-AT proteins using specific antibodies that bind to defined terminal regions; and (ii) determining the presence or absence of a pancreatic cystic tumor by inputting the measured values into a preset discriminant and comparing the resulting value to a reference discriminant. These additional elements likewise do not integrate the judicial exceptions into a practical application. First, the use of antibodies to measure protein levels represents a conventional and well-understood laboratory technique (e.g., immunoassays). The claim does not recite any novel antibody structure, unconventional binding mechanism, or improved assay configuration. As such, this step merely constitutes routine data-gathering activity, which is insufficient to integrate the judicial exception. Second, the step of inputting measured values into a "preset discriminant" and comparing the result represents data processing and analysis, which falls within the abstract idea identified in Step 2A, Prong 1. This step does not recite any specific improvement to mathematical modeling, computational techniques, or data processing technology. Instead, it broadly encompasses applying a decision rule or statistical model to the measured values, which is a conventional analytical activity. Lastly, claim 8 does not recite any particular machine or apparatus that is integral to performing the discriminant analysis, nor does it improve the functioning of any computer or analytical system. The claim also does not effect a transformation of matter beyond routine measurement and evaluation of a sample. Additionally, like claim 1, claim 8 does not apply the judicial exception to effect a treatment or prophylaxis, but merely provides diagnostic information.

Step 2B, Inventive Concept (Refer to MPEP §2106.05): Regarding claim 1, the additional elements in claim 1 consist of: obtaining a biological (body fluid) sample from a subject; measuring the amount of APOA2-AT and/or APOA2-ATQ proteins in the sample; and using the measured values to assist in

determining whether a pancreatic cystic tumor is benign or malignant. These additional steps recited in claim 1 were previously taken by those in the field as demonstrated by Honda et al. (Plasma Biomarker for Detection of Early Stage Pancreatic Cancer and Risk Factors for Pancreatic Malignancy Using Antibodies for Apolipoprotein-AII Isoforms. Scientific Reports. Vol. 5, No. 1, November 2015 – IDS dated 09/21/2023). Honda et al. demonstrates that plasma/serum biomarkers are measured from biological samples and used for detecting pancreatic cancer and assessing risk of pancreatic malignancy, wherein plasma and serum concentrations of apoAII isoforms are measured in cohorts of subjects including healthy individuals and patients with pancreatic cancer. Honda et al. further demonstrates that apoAII isoforms, including apoAII-ATQ and apoAII-AT, are measured using established analytical techniques, such as mass spectrometry-based proteomic analysis and enzyme-linked immunosorbent assays (ELISAs), to evaluate differences between cancer patients and healthy controls (Results, paragraph 1, page 2). Additionally, Honda et al. further demonstrates that these measured biomarker levels are used to identify patients at risk for pancreatic malignancy and to detect early-stage pancreatic cancer (Abstract, page 1). Accordingly, these elements represent well-understood, routine, and conventional activities previously known in the field and do not amount to significantly more than the judicial exception.

Additionally, the additional elements recited in claims 2–8 do not amount to significantly more than the identified judicial exceptions. These claims merely add further details such as specifying particular types of pancreatic cystic tumors, defining comparisons to threshold or relative values, or identifying particular sample types. Such limitations represent routine refinements and conventional variations of the diagnostic process.

Regarding claims 3-6, the additional elements do not amount to significantly more than the identified judicial exceptions. The recited comparison and threshold-based limitations represent well-understood, routine, and conventional data analysis techniques used in the field of biomarker-based diagnostics. These conventional analytical techniques do not provide an inventive concept. Accordingly,

claims 3–6, when considered individually and as an ordered combination, do not recite additional elements that amount to significantly more than the judicial exceptions.

Regarding claim 8, even when considered independently, it recites additional limitations directed to: measuring APOA2-ATQ and APOA2-AT proteins using specific antibodies; and processing the measured values (e.g., inputting into a discriminant and comparing results) to determine the presence or absence of a pancreatic cystic tumor. These limitations remain within the realm of well-understood, routine, and conventional activities. Honda et al. demonstrates that specific antibodies are developed and used in sandwich ELISAs to measure apoAII isoforms, including apoAII-ATQ and apoAII-AT, wherein antibodies specific to these isoforms are used as capture and detection antibodies in immunoassays to quantify biomarker levels in plasma samples (Results, paragraph 1, page 2). Honda et al. further demonstrates that measured biomarker values are analyzed using statistical and computational methods, including calculation of surrogate biomarker values (page 4), receiver operating characteristic (ROC) analyses, and area under the curve (AUC) determinations to classify and distinguish disease states (pages 4-6). These teachings confirm that antibody-based detection of biomarkers and subsequent mathematical/statistical analysis of measured values were conventional techniques in the field.

Accordingly, claims 1-8 are rejected under 35 U.S.C. 101 as being directed to judicial exceptions, including natural correlations and abstract ideas, without reciting additional elements that integrate the exceptions into a practical application or amount to significantly more than the exception that would render the claims patent eligible.

Claim Rejections - 35 USC § 103

In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis (i.e., changing from AIA to pre-AIA) for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

The factual inquiries for establishing a background for determining obviousness under 35 U.S.C. 103 are summarized as follows:

1. Determining the scope and contents of the prior art.

2. Ascertaining the differences between the prior art and the claims at issue.

3. Resolving the level of ordinary skill in the pertinent art.

4. Considering objective evidence present in the application indicating obviousness or nonobviousness.

This application currently names joint inventors. In considering patentability of the claims the examiner presumes that the subject matter of the various claims was commonly owned as of the effective filing date of the claimed invention(s) absent any evidence to the contrary. Applicant is advised of the obligation under 37 CFR 1.56 to point out the inventor and effective filing dates of each claim that was not commonly owned as of the effective filing date of the later invention in order for the examiner to

consider the applicability of 35 U.S.C. 102(b)(2)(C) for any potential 35 U.S.C. 102(a)(2) prior art against the later invention.

For purposes of examination under 35 U.S.C. § 103, the claims are interpreted to be consistent with the scope of enablement. Accordingly, for purposes of prior art analysis, the claims are reasonably interpreted as encompassing methods in which APOA2-AT and/or APOA2-ATQ levels are measured in a body fluid sample and the measured amounts are used to assist in distinguishing benign and malignant pancreatic cystic tumors within the limited context supported by the specification, including IPMN or similar cystic lesions. The phrase "assisting in the determination" encompasses the use of biomarker level differences or comparisons (e.g., relative increases or decreases or comparison to reference values) and does not require a definitive or universally applicable classification rule. Thus, the claims are not interpreted as requiring reliable or exclusive differentiation across all pancreatic cystic tumor types, but rather as encompassing methods that provide biomarker-based information correlated with malignancy status within the scope supported by the disclosure, which demonstrates limited and moderately discriminatory performance.

Claims **1-7** are rejected under 35 U.S.C. 103 as being unpatentable over **Honda et al.** (Plasma Biomarker for Detection of Early Stage Pancreatic Cancer and Risk Factors for Pancreatic Malignancy Using Antibodies for Apolipoprotein-AII Isoforms. Scientific Reports. Vol. 5, No. 1, November 2015 – IDS dated 09/21/2023) in view of **Okusaka et al.** (Altered Plasma Apolipoprotein Modifications in Patients with Pancreatic Cancer: Protein Characterization and Multi-Institutional Validation. PloS One. Vol. 7, No. 10, October 2012 – IDS dated 09/21/2023) and **UniProt** (P02652, https://www.uniprot.org/uniprotkb/P02652/entry), **Talar et al**. (Pancreatic Cyst Fluid Analysis for Differential Diagnosis between Benign and Malignant Lesions. Oncology Letters. Vol. 5, No. 2, February 2013), and **Maire et al.** (Intraductal Papillary Mucinous Neoplasms of the Pancreas: Performance of

Pancreatic Fluid Analysis for Positive Diagnosis and the Prediction of Malignancy. The American Journal of Gastroenterology. Vol. 103, No. 11, November 2008).

Regarding claim **1**, Honda et al. teaches a method for detecting pancreatic malignancy using plasma biomarkers. Specifically, Honda et al. describes that plasma biomarkers are used for screening patients and identifying individuals at risk for pancreatic malignancy and early-stage pancreatic cancer (paragraphs 1 and 3, page 2). Honda et al. further teaches that apoAII isoforms enable detection of early-stage pancreatic cancer and identification of patients at high risk for pancreatic malignancy (Discussion, paragraph 1, page 11). Furthermore, Honda et al. teaches that specific apoAII isoforms include apoAII-ATQ/ATQ, apoAII-ATQ/AT, and apoAII-AT/AT, which are defined by the truncation of varying numbers of amino acids from the C-terminus of the apoAII protein (paragraph 2, page 2). Honda et al. discloses that ELISA assays were developed using antibodies specific to apoAII-AT and apoAII-ATQ isoforms, enabling measurement of these specific isoforms in biological samples (Results, paragraph 1, page 2). Moreover, Honda et al. teaches that plasma samples are collected and analyzed for apoAII isoforms using ELISA and mass spectrometry (Results, paragraph 3, page 2).

Honda et al. further teaches that measured apoAII isoform levels are used in a comparative diagnostic framework to distinguish disease states. Specifically, teaches that levels of apoAII-ATQ/AT were significantly lower in patients with pancreatic cancer compared to healthy controls and are used to distinguish disease states (Results, paragraph 3, page 4; Results, paragraph 2, page 6). Additionally, Honda et al. teaches that receiver operating characteristic (ROC) analyses are performed to evaluate diagnostic performance, wherein the area under the curve (AUC) for apoAII-ATQ/AT determined by ELISA is 0.935, which is higher than that for both apoAII-ATQ and apoAII-AT, thereby demonstrating discriminatory capability between disease and non-disease states (Results, paragraph 5, page 4; Fig. 2C–2F, page 5). Thus, Honda et al. establishes that measured biomarker amounts are not merely detected, but are

quantitatively compared and evaluated using statistical thresholds and performance metrics to assist in diagnostic determinations.

Although Honda et al. teaches measuring apoAII isoforms (apoAII-ATQ/AT) using antibodies and using their levels for diagnosing pancreatic malignancy, Honda et al. does not explicitly teach the specific amino acid sequences recited in claim 1 (SEQ ID NO:30 and SEQ ID NO:31). Hence, Honda et al. lacks explicit sequence identity disclosure.

Okusaka et al. teaches that distinct ApoAII isoforms are defined based on specific C-terminal amino acid compositions, including ApoAII-1 is a homodimer of untruncated ApoAII peptides with C-terminal ends of-ATQ/-ATQ. ApoAII-3 is a homodimer of AII peptide chains, both the C-terminal ends of which lack glutamine (Q) residues (-AT/-AT) (Results, paragraph 2, page 4; Figure 3B). Okusaka et al. further discloses that antibodies can be generated that specifically recognize ApoAII peptides depending on whether the C-terminus contains (-ATQ), or lacks specific residues (-AT) such as glutamine (Results, paragraph 5, page 7), thereby demonstrating that isoform identity is directly tied to defined peptide sequences. This establishes that apoAII isoforms are inherently sequence-defined structures based on specific amino acid truncations.

UniProt, a widely accessible public protein database, provides the full-length ApoA-II sequence (P02652). From the record (refer to figure below), the C-terminal region includes sequence ending in: GTQPATQ. From this sequence, truncation yields to GTQPAT (SEQ ID NO: 30, APOA2-AT) or TQPATQ (SEQ ID NO: 31, APOA2-ATQ). These exact sequences correspond to: SEQ ID NO: 30 (Gly Thr Gln Pro Ala Thr) and SEQ ID NO: 31 (Thr Gln Pro Ala Thr Gln). Additionally, UniProt shows the entry of P02652, released since 11/01/1988. Therefore, the ApoA-II sequence was publicly available since 1988.

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>sp|P02652|Release 9.0|01-Nov-1988
MKLLAATVLLLTICSLEGALVRRQAKEPCVESLVSQYFQTVTDYGKDLMEKVKSPELQAE
AKSYFEKSKEQLTPLIKKAGTELVNFLSYFVELGTQPATQ
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Talar et al. teaches a diagnostic framework for pancreatic cystic tumors in which pancreatic cyst fluid biomarkers are quantitatively measured and used to distinguish benign pancreatic cysts from premalignant or malignant cystic lesions, including intraductal papillary mucinous neoplasm (IPMN). Specifically, Talar et al. discloses that cysts were classified as benign (simple cysts, pseudocysts and serous cystadenomas) in in 36 patients or premalignant/malignant (mucinous cystadenomas, IPMN and cystadenocarcinomas) in 16 patients, and that biochemical marker (e.g., CEA, CA 19-9, and amylase) levels are analyzed to differentiate between these groups (Abstract, page 613). Talar et al. further teaches that biomarker levels are quantitatively measured and compared between these groups, wherein CEA and CA 19-9 levels are elevated in patients with malignant cysts compared with benign lesions with statistically significant differences ($P<0.001$) (Abstract, page 613). Additionally, Talar et al. teaches that cyst fluid analysis is used for the differential diagnosis of pancreatic cystic lesions and assists in distinguishing benign from malignant lesions (Abstract, page 613). Talar et al. further teaches that receiver operating characteristic (ROC) analysis is used to depict the ability to discriminate between benign and premalignant/malignant cysts (Materials and Methods, paragraph 4, page 614).

Maire et al. further reinforces and specifically exemplifies this comparative diagnostic framework within IPMN, consistent with the scope of enablement. Maire et al. teaches measuring tumor markers, including CEA, CA 19.9, and CA 72.4, in pancreatic cyst fluid obtained via endoscopic ultrasonography-guided fine-needle aspiration (EUS-FNA) (Methods, page 2871), and evaluating their diagnostic utility in differentiating benign and malignant intraductal papillary mucinous neoplasms (IPMNs) (Results, page 2871). Maire et al. further teaches that the levels of CEA, CA 19.9, and CA 72.4 were significantly different between benign IPMN and malignant IPMN (Results, page 2871), thereby demonstrating that measured biomarker amounts are used to distinguish benign and malignant states within IPMN. Additionally, Maire et al. teaches that receiver operating characteristic (ROC) curves are used to select cutoff values that maximize the difference between benign and malignant IPMN (Patients and Methods, paragraph 4, page

2873). Maire et al. further teaches that, for example, a CEA cutoff of 200 ng/mL yields sensitivity of 90%, specificity of 71%, and negative predictive value of 96% for distinguishing malignant IPMN, thereby confirming that comparative biomarker levels and threshold-based evaluation are used to assist in determining malignancy status within IPMN populations (Results, paragraph 1, page 2874). Thus, Talar et al. and Maire et al. collectively teach the use of measured biomarker amounts within a comparative framework to assist in distinguishing benign and malignant pancreatic cystic tumors, including IPMN, consistent with the scope of enablement.

Therefore, it would have been obvious to one of ordinary skill in the art before the effective filing date of the claimed invention to modify the diagnostic method of Honda et al., as informed by Okusaka et al. and UniProt for defining and identifying APOA2 isoforms at the sequence level, to further apply the measured amounts of APOA2-AT and/or APOA2-ATQ within a comparative diagnostic framework for distinguishing benign and malignant pancreatic cystic tumors, including IPMN or similar cystic lesions, as taught by Talar et al. and further exemplified and statistically validated by Maire et al., in order to provide clinically meaningful differentiation between benign and malignant cystic lesions.

A person having ordinary skill in the art (PHOSITA) would have recognized that Honda et al. already establishes a quantitative and statistically validated biomarker framework in which apoAII isoform levels are measured and evaluated using comparative analyses, including ROC curves and AUC values, to distinguish between disease and non-disease states. Talar et al. teaches that pancreatic cystic tumors, including IPMN, are classified into benign and malignant groups and that biomarker levels are compared between these groups to assist in diagnosis. Importantly, Talar et al. and Maire et al. demonstrate that this diagnostic approach is not limited to any single specific biomarker, but rather reflects a general diagnostic framework in which measured biomarker levels—regardless of identity—are quantitatively compared between benign and malignant populations to assess disease status. Thus, the prior art establishes that the use of comparative biomarker behavior (e.g., relative increases, decreases, or

threshold-based differences) is a generalizable and well-understood principle for distinguishing benign and malignant pancreatic cystic lesions, including IPMN. Thus, combining Honda et al.'s isoform-based biomarker measurement system with the comparative diagnostic frameworks of Talar et al. and Maire et al. would have represented a predictable use of prior art elements according to their established functions, namely, using measured biomarker amounts and relative differences or threshold-based comparisons to assist in determining malignancy status within pancreatic cystic tumors, including IPMN or similar cystic lesions. This directly corresponds to the claimed "assisting in the determination," as interpreted in light of the scope of enablement. Furthermore, a PHOSITA would have been motivated to make this modification because clinical management of pancreatic cystic lesions, particularly IPMN, depends on distinguishing benign from malignant conditions, and both Talar et al. and Maire et al. demonstrate that quantitative biomarker differences and threshold-based comparisons provide clinically actionable diagnostic information. Given that the prior art teaches that multiple distinct biomarkers (e.g., CEA, CA 19.9, CA 72.4) may be used within the same comparative diagnostic framework, a skilled artisan would have recognized that incorporating an additional known and measurable biomarker—such as APOA2 isoforms already demonstrated by Honda et al. to correlate with pancreatic disease—would have been a routine and predictable extension of this established diagnostic approach. The incorporation of sequence-defined isoform specificity from Okusaka et al. and UniProt would have further improved assay specificity, reproducibility, and molecular definition without altering the underlying comparative diagnostic framework.

Lastly, a PHOSITA would have had a reasonable expectation of success in making this modification because immunoassay-based biomarker measurement, comparative statistical analysis, and ROC-based diagnostic evaluation were well established and routinely applied in pancreatic disease diagnostics. Honda et al. demonstrates successful application of these techniques to apoAII isoforms with strong discriminatory performance, while Maire et al. confirms that similar comparative approaches reliably

distinguish benign and malignant IPMN using defined cutoff values and statistical validation metrics. Accordingly, applying these known analytical and statistical techniques to the APOA2-AT and APOA2-ATQ isoforms within the context of pancreatic cystic tumors (IPMN) would have yielded predictable and reliable diagnostic information, consistent with the level of ordinary skill in the art.

Regarding, claim **2**, Honda et al. teaches that pancreatic diseases analyzed include intraductal papillary mucinous neoplasms (IPMNs) and mucinous cystic neoplasms (MCNs), which are explicitly identified as pancreatic disease categories evaluated in the study population (Results, paragraph 3, page 8). Additionally, Honda et al. teaches that these pancreatic disease types, including IPMNs and MCNs, are considered conditions associated with increased risk for pancreatic malignancy and are evaluated using the same apoAII isoform biomarker framework (Results, paragraph 3, page 8).

Regarding claim **3**, Honda et al. teaches that the average plasma level of apoAII-AT was significantly lower in pancreatic cancer (IDACP) patients compared with healthy controls (Results, paragraph 3, page 4; Fig. 2B, left panel), and that biomarker concentrations can be used to distinguish disease states based on quantitative differences in measured levels. While Talar et al. teaches that certain biomarkers exhibit lower values in malignant cysts compared with benign lesions, such as mean amylase level in benign lesions being significantly higher compared with malignant pancreatic cysts (Results, paragraph 1, page 615). Thus, Talar et al. demonstrates that directional differences in biomarker levels, including cases where malignant values are lower than benign values, are used to assist in distinguishing disease states.

Therefore, it would have been obvious to one of ordinary skill in the art before the effective filing date of the claimed invention to modify the diagnostic method of Honda et al., as further informed by Okusaka et al. and UniProt for defining and identifying the APOA2-AT isoform, to include comparison of measured APOA2-AT levels with values associated with benign pancreatic cystic tumors, as taught by Talar et al., in order to improve the clinical relevance and interpretability of biomarker-based determinations

within pancreatic cystic tumor populations. A PHOSITA would have understood that comparing a measured APOA2-AT level from a test subject to levels associated with benign pancreatic cystic tumors—whether derived from individual samples or representative benign cohorts—constitutes a routine and predictable application of known comparative diagnostic principles within pancreatic cystic tumor populations, including IPMN. Incorporating a known benign cyst comparator would have been understood as a routine refinement of the control population, substituting a more clinically relevant comparator for a general control, thereby improving diagnostic specificity without altering the underlying measurement methodology. Lastly, a PHOSITA would have had a reasonable expectation of success in making this modification because given that both references rely on quantitative biomarker measurement and comparative analysis, and that such comparisons are known to provide correlative diagnostic insight rather than definitive classification, applying Honda et al.'s biomarker within Talar et al.'s benign-versus-malignant comparator framework would have predictably yielded consistent, reproducible, and clinically meaningful information for assisting in distinguishing malignancy within pancreatic cystic tumor populations, including IPMN.

Regarding claim **4**, Honda et al. explicitly teaches defining diagnostic cutoff values for biomarkers, including apoAII-ATQ/AT and CA19-9, and applying those thresholds to classify subjects. Specifically, Honda et al. defined cut-off values for apoAII-ATQ/AT and CA19–9 as 46.3 µg/ml and 75 units/ml, respectively, and further defined an apoAII-ATQ/AT level of less than 46.3 µg/ml and/or a CA19–9 level of more than 75 units/ml as indicative of pancreatic cancer (IDACP) (Results, paragraph 4, page 6). Thus, Honda et al. teaches that biomarker concentration thresholds are established based on observed distributions in patient populations and are used to distinguish disease states. While, Talar et al. likewise teaches that diagnostic biomarkers such as CA19-9 are evaluated using defined cutoff values to distinguish between benign and malignant pancreatic lesions. Specifically, Talar et al. teaches that CA 19-9 levels, with a cut-off value of 37 U/ml, were elevated in patients with malignant cysts compared with benign

lesions. The results are consistent with previous studies reporting that low CA 19-9 fluid levels (less than 37 U/ml) suggest benign lesions (Discussion, paragraph 5, page 615). Talar et al. further teaches that such cutoff thresholds are routinely used in clinical practice and that adjusting the cutoff value directly impacts diagnostic performance, including sensitivity and specificity. In particular, increasing the cutoff value for CA 19-9 to support the diagnosis of a malignant cyst has been previously demonstrated to increase the specificity but decrease the sensitivity of the test. (Discussion, paragraph 5, page 615), confirming that threshold-based interpretation is a known and routine aspect of biomarker-based diagnostics.

Therefore, it would have been obvious to one of ordinary skill in the art before the effective filing date of the claimed invention to modify the diagnostic method of Honda et al., as further informed by Okusaka et al. and UniProt for defining and identifying the APOA2-AT isoform, to employ cutoff values for APOA2-AT biomarker levels derived from known benign and/or malignant pancreatic cystic tumor samples, as further taught by Talar et al., in order to provide a standardized and clinically actionable framework for interpreting biomarker measurements in assisting the distinction between benign and malignant pancreatic cystic lesions. A PHOSITA would have understood that the use of cutoff values derived from known benign or malignant pancreatic cystic tumor samples represents a conventional approach for assisting in the interpretation of biomarker levels, rather than providing a definitive or universally applicable classification rule. Also, a PHOSITA would have recognized that introducing such thresholds provides a practical decision-support tool for interpreting continuous biomarker data within cystic tumor populations, including IPMN. Lastly, a PHOSITA would have had a reasonable expectation of success in making this modification because given that APOA2-AT is already quantitatively measurable using established assays, its concentration correlated with malignancy, threshold-based interpretation is routinely used in pancreatic cyst diagnostics, the application of cutoff values derived from benign and malignant cyst populations would have represented only a routine and inevitable application of known

statistical and diagnostic principles, yielding predictable and clinically meaningful results for assisting in the determination of malignancy within pancreatic cystic tumor populations, including IPMN.

Regarding claim **5**, Honda et al. teaches that the average plasma level of apoAII-ATQ was slightly but significantly higher in IDACP patients compared with healthy controls (Results, paragraph 3, page 4; Fig. 2B, middle panel), thereby demonstrating that APOA2-ATQ biomarker increases in malignant conditions relative to non-malignant subjects. This teaching evidences that APOA2-ATQ levels are elevated in malignant disease states and are associated with malignancy. While, Maire et al. further discloses that median CEA levels are substantially higher in malignant IPMN (5,790 U/mL) compared to benign IPMN (100 U/mL), and similarly, CA 19.9 and CA 72.4 levels are also higher in malignant cases compared to benign cases, demonstrating a consistent pattern of increased biomarker levels in malignant disease (Table 3, page 2873). Accordingly, Maire et al. teaches the principle that malignant pancreatic cystic tumors are associated with higher biomarker levels relative to benign tumors, thereby demonstrating a consistent pattern of increased biomarker levels in malignant pancreatic cystic tumors.

Therefore, it would have been obvious to one of ordinary skill in the art before the effective filing date of the claimed invention to modify the diagnostic method of Honda et al., as further informed by Okusaka et al. and UniProt for defining and identifying the APOA2-AT isoform, to assist in determining malignancy of pancreatic cystic tumors by comparing the measured amount of APOA2-ATQ protein in a test subject to that of a subject having a known benign pancreatic cystic tumor, wherein higher levels indicate malignancy, as taught by Maire et al., in order to align the biomarker analysis with clinically relevant differentiation between benign and malignant cystic lesions. A subject having a known benign pancreatic cystic tumor, as recited in claim 5, represents a refined, disease-relevant control population, and comparing biomarker levels of a test subject to such a benign reference constitutes a routine and well-established comparative diagnostic framework. Lastly, a PHOSITA would have had a reasonable expectation of success in making this modification because both references demonstrate that increased

biomarker levels correlate with malignant disease states, applying this known "increase-based" comparative framework to APOA2-ATQ represents no more than the predictable use of prior art elements according to their established functions. Furthermore, the selection of biomarker directionality (increase versus decrease) constitutes routine optimization of a result-effective variable and does not require inventive skill.

Regarding claim **6**, as discussed above, Honda et al. a teaches defining diagnostic cutoff values for biomarkers, including apoAII-ATQ, and applying those thresholds to classify subjects and diagnose pancreatic cancer. (Results, paragraph 4, page 6). While, Maire et al. teaches that tumor marker levels in pancreatic cyst fluid are quantitatively measured and compared between benign and malignant IPMN, and that cutoff values are established to distinguish between these disease states. Specifically, Maire et al. discloses that by using receiver-operator characteristic (ROC) curves, cutoffs were selected to maximize the difference between benign and malignant IPMN (Patients and Methods, paragraph 4, page 2873). Furthermore, Maire et al. discloses specific cutoff values are defined for tumor markers, such as a CEA level greater than 200 ng/mL and a CA 72.4 level greater than 40 U/mL, and that these thresholds are used to diagnose malignant IPMN and distinguish malignant from benign lesions based on measured biomarker levels (Results, paragraphs 1-2, page 2874, Table 4). Thus, Maire et al. teaches a diagnostic framework in which measured biomarker levels are compared against cutoff values derived from known benign and malignant populations, and wherein values above the cutoff are indicative of malignancy.

Therefore, it would have been obvious to one of ordinary skill in the art before the effective filing date of the claimed invention to modify the diagnostic method of Honda et al., as further informed by Okusaka et al. and UniProt for defining and identifying the APOA2-AT isoform, to include determining that malignancy is assisted when the amount of APOA2-ATQ protein exceeds a cutoff value derived from known benign or malignant populations, as taught by Maire et al., in order to provide standardized, clinically actionable decision criteria and improve diagnostic interpretability. Incorporating cutoff

thresholds into the modified method of Honda et al. represents nothing more than formalizing an inherent and necessary step for translating continuous biomarker measurements into clinically interpretable classifications. The application of such thresholds is not optional, but rather constitutes a conventional and predictable analytical step routinely employed in biomarker-based diagnostics. Lastly, a PHOSITA would have had a reasonable expectation of success in making this modification because both Honda et al. and Maire et al. demonstrate that biomarker levels correlate reliably with disease state and that diagnostic performance can be optimized using defined thresholds. The use of ROC-derived cutoff values and quantitative immunoassay techniques (e.g., ELISA) was well-established, standardized, and routinely used to detect differences in protein levels across patient populations. Therefore, the modification would have involved only routine application of known statistical and diagnostic techniques, yielding predictable and reliable results without undue experimentation.

Regarding claim **7**, Honda et al. teaches that plasma and serum concentrations of apoAII-ATQ/AT are measured in patient cohorts including healthy controls and patients with pancreatic cancer and gastroenterologic diseases (Abstract, page 1). Hence, Honda et al. explicitly teaches the use of plasma and serum samples, both of which are well-known components derived from blood and routinely used in biomarker-based diagnostic assays

Claim **8** is rejected under 35 U.S.C. 103 as being unpatentable over **Honda et al.**, **Okusaka et al.**, and **UniPort**, **Talar et al.,** and **Maire et al.,** as applied to claim 1 above, and further in view of **Sanada et al**. (EP3054298B1) and **Van Regenmortel et al**. (Mapping Epitope Structure and Activity: From One Dimensional Prediction to Four-Dimensional Description of Antigenic Specificity. Methods. Vol. 9, No. 3, June 1996).

With respect to the teachings of Honda et al., Okusaka et al., and UniProt, Talar et al., and Maire et al., see the discussion above, which applies equally here. These references differ from the instant claims

in failing to collectively teach or specify that the antibodies used for measuring APOA2-ATQ and APOA2-AT proteins specifically bind to a carboxyl-terminal region consisting of a defined amino acid sequence (SEQ ID NO: 2 and SEQ ID NO: 1), nor antibodies binding to the amino acid sequence other than the carboxyl-terminal region; nor determining the presence or absence of pancreatic cystic tumor by inputting, to a preset discriminant.

However, Sanada et al. teaches a method for detecting pancreatic tumors by measuring the amounts of APOA2 isoforms, including APOA2-AT and APOA2-ATQ, using antibodies that specifically bind to defined C-terminal peptide regions corresponding to specific amino acid sequences. In particular, Sanada et al. expressly discloses that pancreatic tumor detection is performed by: (i) measuring the amount of APOA2-ATQ protein in the sample using an anti-APOA2-ATQ terminus antibody that specifically binds to a C-terminal region of the APOA2-ATQ protein comprising the amino acid sequence represented by SEQ ID NO: 1, and an anti-APOA2-ATQ non-terminus antibody binding to the amino acid sequence other than the C-terminal region; (ii) measuring the amount of APOA2-AT protein using an anti-APOA2-AT terminus antibody that specifically binds to a C-terminal region of the APOA2-AT protein comprising the amino acid sequence represented by SEQ ID NO: 2, and an anti-APOA2-AT non-terminus antibody binding to the amino acid sequence other than the C-terminal region; and (iii) and determining the test subject to have a pancreatic tumor when the resulting discriminant value of the test subject is statistically significantly different as compared with the measurement value or the discriminant value of a normal subject (paragraph [0017], page 5). Additionally, Sanada et al. explicitly teaches antibodies that bind C-terminal regions comprising SEQ ID NO: 1 and SEQ ID NO: 2 (page 30), and the use of those specific sequences as determinants of isoform-specific antibody recognition and detection.

Although Sanada et al. directly teaches antibody binding to C-terminal regions defined by the same sequences recited in claim 8, Sanada et al. does not explicitly state that the antibody binds to a

region “consisting of” only the minimal recited sequence, as opposed to a larger peptide region encompassing those residues.

Van Regenmortel et al. teaches that continuous epitopes correspond to short peptide fragments of a few amino acid residues that can bind to antibodies raised against the intact protein, and explains that such epitopes are operationally defined as short sequences that retain antigenic binding capability (pages 465-466). Van Regenmortel et al. further teaches that when a short sequence of three to six residues is found to bind to an antibody, that peptide is considered a continuous epitope, even if it corresponds to a region within a larger protein structure (paragraph 1, page 466). Van Regenmortel et al. additionally discloses that such studies have shown that only three to five residues of the structural epitope contribute significantly to the binding energy (paragraph 5, page 466), thereby establishing that the functional antibody-binding region is substantially smaller than the full structural region. Lastly, Van Regenmortel et al. discloses that although the proportion of elicited antibodies able to react with short peptides is small, these antibodies are of considerable practical importance since they have many applications as immunological reagents in diagnostics (paragraph 3, page 466). Thus, Van Regenmortel et al. teaches that antibody binding can be achieved using minimal peptide sequences corresponding to short contiguous amino acid segments, thereby supporting the use of peptide sequences consisting only of the recited residues as functional epitopes.

Therefore, it would have been obvious to one of ordinary skill in the art before the effective filing date of the claimed invention to modify the diagnostic method of Honda et al., as already informed by Okusaka et al. and UniProt for defining and identifying APOA2 isoforms at the sequence level, to employ antibodies that bind specifically to the C-terminal peptide regions of APOA2-AT and APOA2-ATQ defined by SEQ ID NO: 1 and SEQ ID NO: 2, as expressly taught by Sanada et al. and further to refine such antibodies to bind to the minimal peptide sequences consisting of those amino acid sequences as functional epitopes in view of Van Regenmortel et al. A PHOSITA would have recognized that Sanada et al. not only identifies

the C-terminal sequences as the distinguishing structural features of the APOA2 isoforms, but also relies on those sequences as the basis for selective antibody recognition and quantitative immunoassay detection, thereby establishing that isoform specificity is inherently tied to the precise amino acid composition of the terminal residues. In view of Van Regenmortel et al.'s teaching that antibody–antigen interactions for linear epitopes are governed by a limited number of contiguous residues and that only a subset of residues within a larger region contributes meaningfully to binding affinity, it would have been an obvious and routine optimization to define and utilize the minimal contiguous peptide sequence corresponding to the functional epitope itself, rather than a larger peptide segment encompassing that sequence. Such refinement would have been motivated by well-established goals in immunoassay design, including increasing binding specificity, minimizing cross-reactivity among closely related isoforms that differ by only one or a few terminal residues, improving signal-to-noise ratio, and enhancing assay reproducibility across different sample sets and antibody preparations. Accordingly, the modification represents the application of known epitope-mapping principles to a known biomarker system to achieve predictable improvements in assay performance.

Lastly, a PHOSITA would have had a reasonable expectation of success in making this modification because given that the APOA2-AT and APOA2-ATQ isoforms differ only at their C-terminal ends and that Sanada et al. already relies on these terminal differences for selective detection, a skilled artisan would have reasonably expected that restricting the antibody-binding region to the minimal sequence consisting of SEQ ID NO: 1 and SEQ ID NO: 2 would preserve the necessary binding interactions while further enhancing selectivity between isoforms. Additionally, techniques for synthesizing short peptides, generating antibodies against such peptides, and incorporating those antibodies into ELISA-based detection systems were well established, standardized, and widely practiced in the art prior to the effective filing date. Thus, the modification would have involved only routine application of established

immunological and biochemical principles, yielding predictable improvements in assay specificity and performance without undue experimentation.

Ultimately, claims 1–8 are rejected under 35 U.S.C. 103 as being unpatentable over Honda et al. in view of Okusaka et al. and UniProt, and further in view of Talar et al., Maire et al., Sanada et al., and Van Regenmortel et al. As discussed above, the claims are interpreted to be consistent with the scope of enablement. The cited references collectively teach and suggest measuring APOA2 isoforms, correlating biomarker levels with disease state, performing comparative and threshold-based analyses, and employing sequence-specific antibody detection targeting C-terminal regions of APOA2 isoforms. The additional limitations, including relative comparisons, cutoff-based determinations, and minimal epitope-based antibody specificity, represent no more than routine optimization of result-effective variables and application of well-established biochemical, immunological, and diagnostic principles. A skilled artisan would have been motivated to combine these teachings and would have had a reasonable expectation of success in doing so, yielding predictable and clinically meaningful results. Therefore, the claimed subject matter as a whole would have been obvious before the effective filing date of the claimed invention.

Double Patenting

The nonstatutory double patenting rejection is based on a judicially created doctrine grounded in public policy (a policy reflected in the statute) so as to prevent the unjustified or improper timewise extension of the "right to exclude" granted by a patent and to prevent possible harassment by multiple assignees. A nonstatutory double patenting rejection is appropriate where the conflicting claims are not identical, but at least one examined application claim is not patentably distinct from the reference claim(s) because the examined application claim is either anticipated by, or would have been obvious over, the reference claim(s). See, e.g., *In re Berg*, 140 F.3d 1428, 46 USPQ2d 1226 (Fed. Cir. 1998); *In re Goodman*,

11 F.3d 1046, 29 USPQ2d 2010 (Fed. Cir. 1993); *In re Longi*, 759 F.2d 887, 225 USPQ 645 (Fed. Cir. 1985); *In re Van Ornum*, 686 F.2d 937, 214 USPQ 761 (CCPA 1982); *In re Vogel*, 422 F.2d 438, 164 USPQ 619 (CCPA 1970); *In re Thorington*, 418 F.2d 528, 163 USPQ 644 (CCPA 1969).

A timely filed terminal disclaimer in compliance with 37 CFR 1.321(c) or 1.321(d) may be used to overcome an actual or provisional rejection based on nonstatutory double patenting provided the reference application or patent either is shown to be commonly owned with the examined application, or claims an invention made as a result of activities undertaken within the scope of a joint research agreement. See MPEP § 717.02 for applications subject to examination under the first inventor to file provisions of the AIA as explained in MPEP § 2159. See MPEP § 2146 *et seq.* for applications not subject to examination under the first inventor to file provisions of the AIA. A terminal disclaimer must be signed in compliance with 37 CFR 1.321(b).

The filing of a terminal disclaimer by itself is not a complete reply to a nonstatutory double patenting (NSDP) rejection. A complete reply requires that the terminal disclaimer be accompanied by a reply requesting reconsideration of the prior Office action. Even where the NSDP rejection is provisional the reply must be complete. See MPEP § 804, subsection I.B.1. For a reply to a non-final Office action, see 37 CFR 1.111(a). For a reply to final Office action, see 37 CFR 1.113(c). A request for reconsideration while not provided for in 37 CFR 1.113(c) may be filed after final for consideration. See MPEP §§ 706.07(e) and 714.13.

The USPTO Internet website contains terminal disclaimer forms which may be used. Please visit www.uspto.gov/patent/patents-forms. The actual filing date of the application in which the form is filed determines what form (e.g., PTO/SB/25, PTO/SB/26, PTO/AIA/25, or PTO/AIA/26) should be used. A web-based eTerminal Disclaimer may be filled out completely online using web-screens. An eTerminal Disclaimer that meets all requirements is auto-processed and approved immediately upon submission.

For more information about eTerminal Disclaimers, refer to www.uspto.gov/patents/apply/applying-online/eterminal-disclaimer.

As discussed above, for purposes of this rejection, the instant claims are interpreted consistent with the scope of enablement.

Claims **1-8** are rejected on the ground of nonstatutory double patenting as being unpatentable over claim 1-6 of U.S. Patent No. **11,162,956** in view of Honda et al. Okusaka et al., UniProt entry P02652, Talar et al., Maire et al., Sanada et al., and Van Regenmortel et al.

Patent claim 1 teaches a method for detecting disturbance of pancreatic juice flow by measuring APOA2-ATQ protein in a body fluid using terminus and non-terminus antibodies and comparing the amount to a control sample. This corresponds to the APOA2 isoform-based diagnostic framework of instant claim 1. However, patent claim 1 does not expressly teach determining malignant pancreatic cystic tumor, defining APOA2-AT and APOA2-ATQ by SEQ ID NOs: 30 and 31, nor applying the measurements within a comparative framework directed to assisting in distinguishing benign and malignant pancreatic cystic tumors within the limited IPMN-type context. As discussed above, Honda et al. teaches that apoAII isoforms are used as plasma biomarkers for screening patients for the early-stage of pancreatic cancer and identifying patients at high risk for pancreatic malignancy. Okusaka et al. teaches that the relevant apoAII isoforms are structurally defined by specific C-terminal truncations, and UniProt entry P02652 provides the full-length ApoA-II sequence from which the claimed APOA2-AT and APOA2-ATQ terminal sequences are directly derivable. Talar et al. and Maire et al. teach that biomarker levels are quantitatively compared between benign and malignant pancreatic cystic tumors, including IPMN, using comparative and threshold-based frameworks to assist in diagnosis. Collectively, these references would have suggested modifying patent claim 1 to define the isoforms at the sequence level and to apply the same biomarker measurement framework within a benign-versus-malignant pancreatic cystic tumor context (IPMN), as a predictable use of known biomarker analysis techniques.

Instant claim 2 is not patentably distinct from patent claim 1. While patent claim 1 broadly applies to pancreatic disorder and/or disease, Honda et al., Talar et al., and Maire et al. teach that pancreatic cystic tumors, including IPMN, are evaluated using biomarker-based comparative frameworks. Applying the method of patent claim 1 to these specific cystic tumor subtypes would have been a predictable use of the same diagnostic system in closely related pancreatic disease contexts.

Instant claims 3-6 are not patentably distinct from patent claims 3–6. The patent claims teach threshold-based comparison of APOA2-ATQ relative to normal subjects but do not expressly recite the benign-versus-malignant cyst comparator framework or the specific lower-than or higher-than relationships recited. As discussed above, Honda et al. teaches that APOA2 isoform levels correlate with malignancy, including decreased APOA2-AT and increased APOA2-ATQ in malignant conditions. Talar et al. teaches comparison of biomarker levels and cutoff values to distinguish benign and malignant pancreatic cysts, including cases where malignant values are lower. Maire et al. further teaches that biomarker levels may be higher in malignant cystic lesions and that ROC-derived thresholds are used to distinguish benign and malignant IPMN. These teachings would have suggested modifying patent claims 3–6 to apply APOA2 biomarker comparisons and threshold-based determinations within a benign-versus-malignant pancreatic cystic tumor framework, including using lower APOA2-AT and higher APOA2-ATQ levels or corresponding cutoff values as indicators of malignancy, as predictable refinements of known biomarker diagnostic principles.

Instant claim 7 is not patentably distinct from patent claim 2. The recitation of blood, plasma, or serum represents routine sample selection for implementing the disclosed APOA2 biomarker assay.

Instant claim 8 is not patentably distinct from patent claim 1. Patent claim 1 teaches antibody-based APOA2 detection using terminus and non-terminus antibodies but does not expressly teach the dual ATQ/AT measurement format with minimal “consisting of” C-terminal sequences or discriminant-based tumor determination. As discussed above, Sanada et al. teaches dual isoform immunoassays with

terminus antibodies and discriminant analysis, and Van Regenmortel et al. teaches that minimal contiguous peptide sequences define functional epitopes. These references would have suggested refining the antibody binding regions the patent's method to the minimal contiguous sequences corresponding to the C-terminal isoform regions, as a predictable application of known epitope-mapping principles to improve specificity in a known APOA2 immunoassay system.

Accordingly, claims 1–8 are not patentably distinct from claims 1–6 of U.S. Patent No. 11,162,956 in view of Honda et al. Okusaka et al., UniProt entry P02652, Talar et al., Maire et al., Sanada et al., and Van Regenmortel et al.

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to ELIZABETH OGUNTADE whose telephone number is (571)272-6802. The examiner can normally be reached Monday-Friday 6:00 AM - 3 PM.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Bao-Thuy Nguyen can be reached at 571-272-0824. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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